

Figure 1: Types of testing

Popular tools for non-functional automation

Product	Vendor
Load Runner	HP
Jmeter	Apache
Burp Suite	PortSwigger
Acunetix	Acunetix

Selenium

Selenium is an open source tool for testing Web apps. It was initially created by ThoughtWorks and now boasts of a dynamic group of designers and clients.

A sample test case

The Selenium IDE is available under *Tools*, in the *Menu* bar of the Firefox browser, as an add-on (see Figure 2).

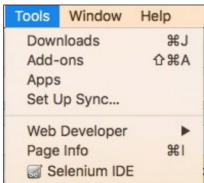


Figure 2: Firefox Menu bar

A window of the Selenium IDE pops up with a blank table of a test case (Figure 3).

As we can see in the window, there is a red dot in the top right corner, which helps in recording the test case. The *Play* options are also available in the top left corner to run the test case. The Base URL talks about the link on which the test case should be applied. We can also slow down or hasten up the testing process. The login at the bottom gives the log information, step wise, about

the success or failure of the test case.

As we can see in this example, we will demonstrate the usage of the Selenium IDE with the Google sign-in process. The input given as the username is *info.epos.project* and the password is *info_epos_* project. Keep the 'stay signed in' option off and try to access the account while recording the test case in the background.

Just after the sign-in, we can also verify the user name present in the location shown in Figure 4, to verify whether the login of the correct user account has taken place or not. This can be done by selecting the name (in the above example, the text is *-EPOS*), right-clicking on the selected content and then clicking on the 'verifyText' (or it could be 'verifyTextPresent' in your version of Selenium) option from the menu. Here, CSS is used to identify the element although ID is more preferred as it would be unique throughout the page. Then sign out. With this, one phase of the test case - the recording - is over.

Once the recording is complete, in the table section of the Selenium IDE, the *Command* column displays the commands (or the actions, to be more precise) which were performed while recording the test case. The *Target* column displays the ID of the element we have dealt with in the Base URL mentioned. Element ID is unique for each element in a single page. Values mentioned here define the values that the ID must contain to get a successful test case.

At the point when the

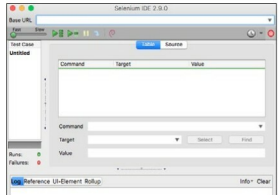


Figure 3: IDE info

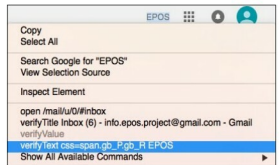


Figure 4: verifyText example

test is running, every command is highlighted on execution. The colour state of the steps turns green or red, to demonstrate achievement or failure, respectively. The dark green shading is for the confirmed content.

After the entire test is finished, the test is stamped as red or green in the suite. The test also generates the log information about the success or failure of the command, with an error message.

Red indicates failure and green indicates success.

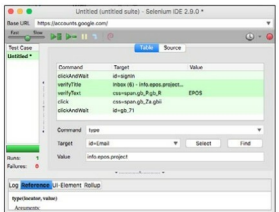


Figure 5: Successful test case